corrigé d° Examen CL (Session juin 2022 Contrôle)

Exercice 1 : (4 pts)

1/ Mux 8 → 1 1pt

2/

	A	B	C	D	S
D_0 = D	0	0	0	0	0
	0	0	0	1	1
D_1 = D	0	0	1	0	1
	0	0	1	1	0
D_2 = D	0	1	0	0	1
	0	1	0	1	0
D_3 = D	0	1	1	0	0
	0	1	1	1	1
D_4 = D	1	0	0	0	1
	1	0	0	1	0
D_5 = D	1	0	1	0	0
	1	0	1	1	1
D_6 = D	1	1	0	0	0
	1	1	0	1	1
D_7 = D	1	1	1	0	1
	1	1	1	1	0

2pts

3/ S !?

AB \ CD	00	01	11	10
00	0	(1)	0	(1)
01	(1)	0	(1)	0
11	0	(1)	0	(1)
10	(1)	0	(1)	0

$S = \bar{A}.\bar{B}.\bar{C}.D + \bar{A}.\bar{B}.C.\bar{D}$
$+ \bar{A}.B(\bar{C}.\bar{D} + C.D) + AB.(\bar{C}.D + C.\bar{D})$
$+ A.\bar{B}.(\bar{A}.\bar{B} + A.B)$ 1pt

$= \bar{A}.\bar{B}.(C \oplus D) + A.B(C \oplus D)$
$+ \bar{A}.B(\overline{C \oplus D}) + A.\bar{B}.(\overline{C \oplus D})$
$= (\overline{A \oplus B}).(C \oplus D) + (A \oplus B).(\overline{C \oplus D})$
$= A \oplus B \oplus C \oplus D$

Exercice 2 (4 pts)

1/

	a	b	c	F
Mem	0	0	0	1
	0	0	1	1
Mise "0"	0	1	0	0
	0	1	1	1
Mise "1"	1	0	0	0
	1	0	1	1
Chg	1	1	0	0
	1	1	1	0

2pts

a_1	a_0	F
0	0	1
0	1	1
1	0	0
1	1	0

2/

a \ bc	00	01	11	10
0	1	1	1	0
1	0	1	0	0

2pts

$F = \bar{a}(\bar{b} + c) + \bar{b}.c$

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Exercice 3 (8 pts)

3 - 4 - 5 - 6 - 7 - 3

1/ Il faut 3 bascules (1 pt)

2/

Q_2	Q_1	Q_0	Q_2	Q_1	Q_0	J_2	K_2	J_1	K_1	J_0	K_0
0	1	1	1	0	0	1	X	X	1	X	1
1	0	0	1	0	1	X	0	0	X	1	X
1	0	1	1	1	0	X	0	1	X	X	1
1	1	0	1	1	1	X	0	X	0	1	X
1	1	1	0	1	1	X	1	X	0	X	0

2 pts

J	K	Q^-	Q
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

⟹

Q^-	Q	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

3/ T.V ⇒ $J_2 = J_0 = 1$

3 pts

Q_2 \ Q_1Q_0	00	01	11	10
0	φ	φ	X	φ
1	0	0	1	0

$K_2 = Q_1 \cdot Q_0$

Q_2 \ Q_1Q_0	00	01	11	10
0	φ	φ	X	φ
1	0	1	X	X

$J_1 = Q_0$

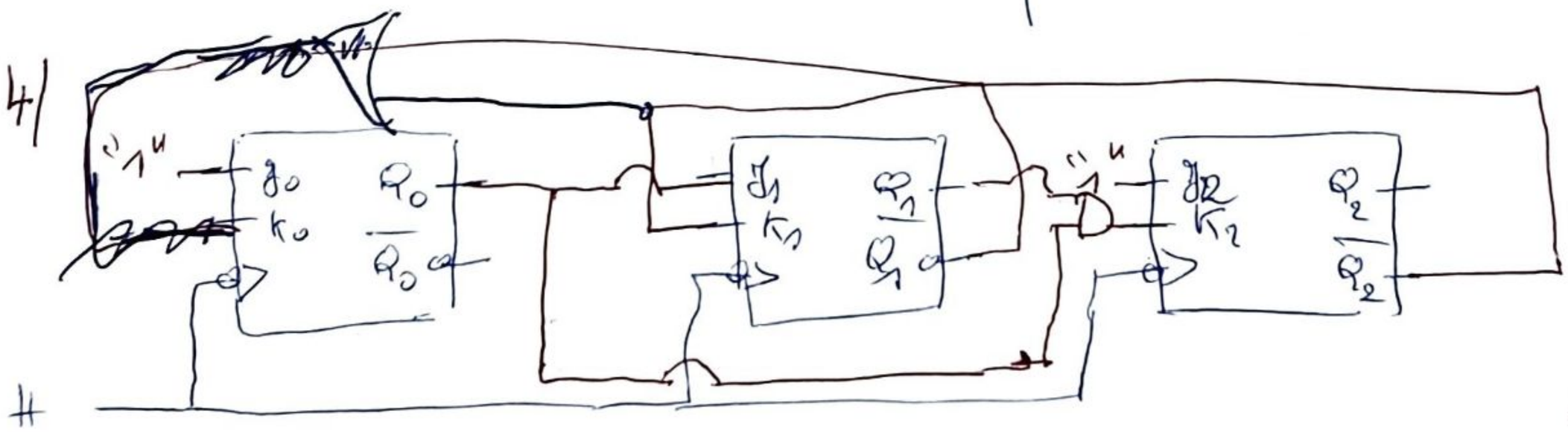
Q_2 \ Q_1Q_0	00	01	11	10
0	φ	φ	1	φ
1	X	X	0	0

$K_1 = \overline{Q_2}$

Q_2 \ Q_1Q_0	00	01	11	10
0	φ	φ	1	φ
1	X	1	0	X

$K_0 = \overline{Q_1} + \overline{Q_2}$

4/



2 pts

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Exercice 4 (4pts)

1/ $y_0 = 1$

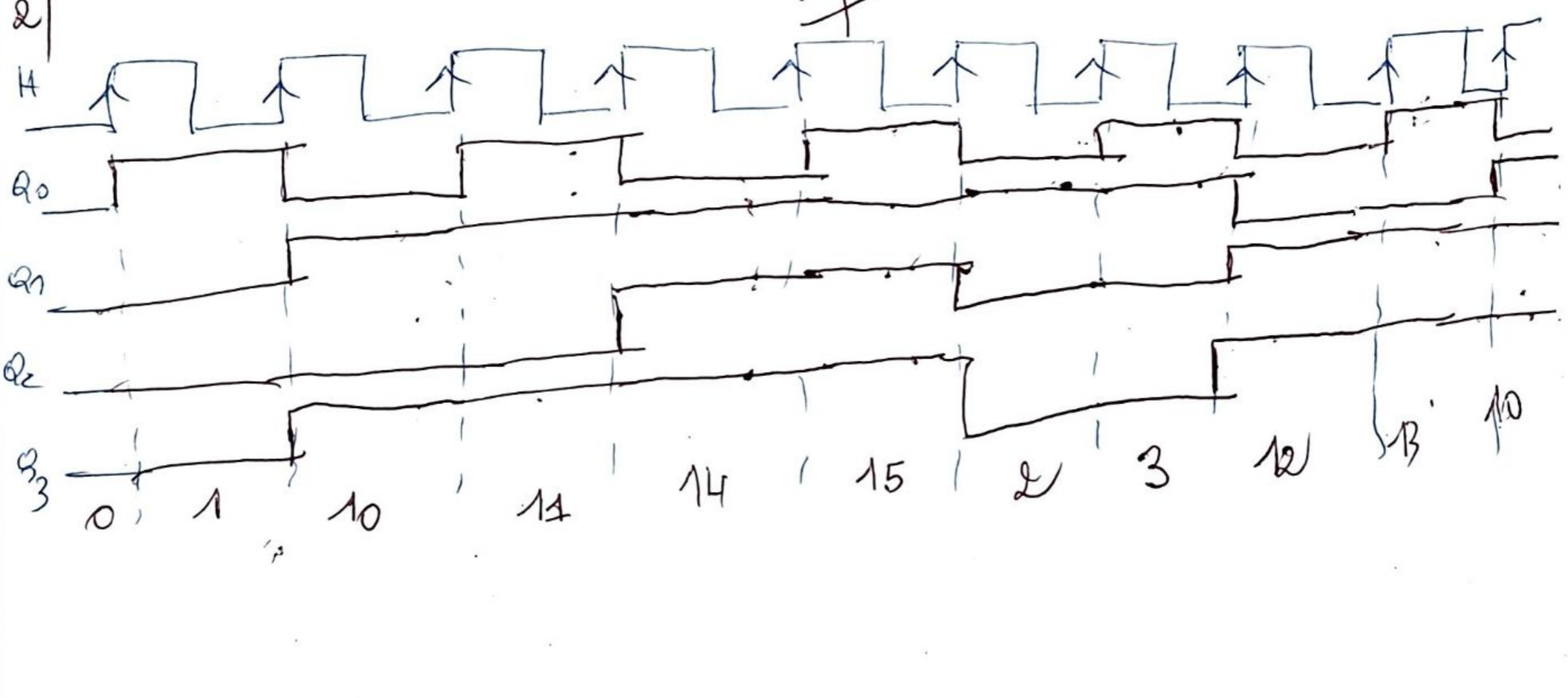
$k_0 = 1$

2pts

$y_1 = Q_0 \cdot \bar{Q}_3$	0	1	0	0	0	0	0	1	0	0
			0	1	0	1	0	1	0	1
$k_1 = Q_0$	0	1	0	1	0	1	0	1	0	0
$y_2 = Q_0 \cdot Q_1$	0	0	0	1	0	1	0	1	0	0
$k_2 = Q_0 \cdot Q_1$	0	0	0	1	0	1	0	0	0	0
$y_3 = Q_0 \cdot Q_1 \cdot Q_2$	0	0	0	0	0	1	0	1	0	1
$k_3 = Q_0$	0	1	0	1	0	1	0	1	0	1

2pts

2/



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